

Ten functional CLTs, written out

Each answer follows the same seven steps: [1] decompose · [2] fix the domain · [3] cite and assemble the derivative · [4] estimator lies in the domain · [5] weak-convergence input · [6] apply Theorem 1.17 · [7] covariance. A closing sanity check follows each proof. Throughout, $q_u = F^{-1}(u)$, B is a standard Brownian bridge on $[0, 1]$ with $\text{Cov}(B(u), B(v)) = u \wedge v - uv$, and $\mathbb{G}$ denotes a generic F -Brownian bridge indexed by a Donsker class. These are written at exam length: every hypothesis is used, nothing is proved that a cited result already gives.

1. Cumulative hazard

Lemma 1.10(2), Theorem 1.22

$\Lambda = -\log(1 - F)$, $F(\tau) < 1$. Functional CLT for $\sqrt{n}(\Lambda_{(n)} - \Lambda)$ on $[0, \tau]$ with covariance.

Theorem. Let $Z_1, \dots, Z_n$ be iid with continuous df F and fix τ with $F(\tau) < 1$. Then

$$\sqrt{n}(\Lambda_{(n)} - \Lambda) \rightsquigarrow \frac{B \circ F}{1 - F} \quad \text{in } \ell^\infty([0, \tau]),$$

a centred Gaussian process with $c(s, t) = \frac{F(s \wedge t) - F(s)F(t)}{(1 - F(s))(1 - F(t))}$.

Proof. [1] $\Lambda = \phi(F)$, $\Lambda_{(n)} = \phi(F_{(n)})$, where $\phi(h) = \varphi \circ h$ is the pointwise map built from $\varphi(x) = -\log(1 - x)$. One stage; no chain rule.

[2] φ' is singular only at 1. Since F is non-decreasing, $\sup_{[0, \tau]} F = F(\tau) < 1$. Put $\delta = \frac{1}{2}(1 - F(\tau))$, so $F(\tau) = 1 - 2\delta$, and $D_\phi = \{h \in \ell^\infty([0, \tau]) : h([0, \tau]) \subseteq (-1, 1 - \delta)\}$, open in $\|\cdot\|_\infty$.

[3] On $(-1, 1 - \delta)$,

$$|\varphi'(x)| = \frac{1}{1 - x} \leq \frac{1}{\delta}, \quad |\varphi''(x)| = \frac{1}{(1 - x)^2} \leq \frac{1}{\delta^2},$$

so φ' is bounded and, being Lipschitz by the mean value theorem, uniformly continuous. Lemma 1.10(2) applies: ϕ is Hadamard differentiable at F tangentially to all of $\ell^\infty([0, \tau])$, with $\phi'_F(h) = h/(1 - F)$, linear with $\|\phi'_F(h)\|_\infty \leq \delta^{-1}\|h\|_\infty$, hence continuous. Since D_ϕ is open and $\|t_n h_n\|_\infty \rightarrow 0$, the perturbations $F + t_n h_n$ lie in D_ϕ for large n .

[4] $F_{(n)}$ is non-decreasing, so $\sup_{[0, \tau]} F_{(n)} = F_{(n)}(\tau) \rightarrow F(\tau) = 1 - 2\delta$ a.s. by Glivenko–Cantelli. Hence $\mathbb{P}(F_{(n)} \in D_\phi) \rightarrow 1$ and $\Lambda_{(n)}$ is well defined on $[0, \tau]$ with probability tending to one.

[5] Theorem 1.22: $\sqrt{n}(F_{(n)} - F) \rightsquigarrow B \circ F$ in $\ell^\infty([0, \tau])$, tight. Only one process enters, so no joint convergence is needed.

[6] Theorem 1.17 with $r_n = \sqrt{n}$: $\sqrt{n}(\Lambda_{(n)} - \Lambda) \rightsquigarrow \phi'_F(B \circ F) = (B \circ F)/(1 - F)$, Gaussian because ϕ'_F is linear.

[7] By monotonicity of F , $F(s) \wedge F(t) = F(s \wedge t)$, so

$$\text{Cov}\left(\frac{B(F(s))}{1 - F(s)}, \frac{B(F(t))}{1 - F(t)}\right) = \frac{F(s) \wedge F(t) - F(s)F(t)}{(1 - F(s))(1 - F(t))} = c(s, t). \quad \square$$

Check. $s = t$ gives $F(t)/(1 - F(t))$: the ordinary delta-method variance of $-\log(1 - \hat{p})$ for a binomial proportion, and Theorem 1.12. Note δ has disappeared.

2. L-statistic $T(F) = \int_0^1 w(u)F^{\leftarrow}(u) du$

Lemma 1.10(1)(i) + Proposition 1.2(1)

w Lipschitz, F continuous with $f > 0$ on an interval containing $\text{supp } w$.

Theorem. Let $[u_1, u_2] = \text{supp } w \subset\subset (0, 1)$, let f be continuous and strictly positive on $[a, b] = [q_{u_1} - \varepsilon, q_{u_2} + \varepsilon]$. Then $\sqrt{n}(T(F_{(n)}) - T(F)) \rightsquigarrow N(0, \sigma^2)$ with

$$\sigma^2 = \iint \frac{w(u)w(v)(u \wedge v - uv)}{f(q_u)f(q_v)} du dv.$$

Proof. [1] $T = \phi_2 \circ \phi_1$ with $\phi_1(F) = F^{\leftarrow} \in \ell^\infty([u_1, u_2])$ and $\phi_2(q) = \int w q du \in \mathbb{R}$.

[2] Stage 1 is localised in the index set, not in the value range: Lemma 1.10(1)(i) needs a compact $[u_1, u_2] \subset (0, 1)$, which $\text{supp } w$ supplies, and f continuous and positive on $[a, b]$. The domain is the set of restrictions of dfs to $[a, b]$; there is no (a, b) -style value cut and no δ . Stage 2 needs no domain at all.

[3] Lemma 1.10(1)(i): $\phi'_{1F}(h) = -(h/f) \circ F^{-1}$, tangentially to $C[a, b]$. For stage 2, $|\phi_2(q)| \leq \|w\|_1 \|q\|_\infty$, so ϕ_2 is bounded linear and by Proposition 1.2(1) is its own derivative, tangentially to everything; in particular the image $\phi'_1(C[a, b])$ lands in its tangential set, so the chain rule (Theorem 1.16) applies:

$$T'_F(h) = - \int_0^1 \frac{w(u)h(q_u)}{f(q_u)} du.$$

[4] $F_{(n)}$ is a df for every n , hence in the domain automatically. Nothing to check.

[5] Theorem 1.22 restricted to $[a, b]$: $\sqrt{n}(F_{(n)} - F) \rightsquigarrow B \circ F$, tight, with continuous paths since F is continuous, so the limit lies in $D_0 = C[a, b]$.

[6] Theorem 1.17, using $F(q_u) = u$:

$$\sqrt{n}(T(F_{(n)}) - T(F)) \rightsquigarrow - \int_0^1 \frac{w(u)B(F(q_u))}{f(q_u)} du = - \int_0^1 \frac{w(u)B(u)}{f(q_u)} du.$$

[7] This is a bounded linear functional of B , hence centred normal, with

$$\sigma^2 = \iint \frac{w(u)w(v)}{f(q_u)f(q_v)} \text{Cov}(B(u), B(v)) du dv = \iint \frac{w(u)w(v)(u \wedge v - uv)}{f(q_u)f(q_v)} du dv. \quad \square$$

Check. Take $w \equiv 1$ (formally part (ii) of the lemma, since $\text{supp } w$ is then not compact in $(0, 1)$), so $T(F) = \mathbb{E}Z$. Substituting $u = F(x)$, $v = F(y)$, $du = f(x)dx$:

$$\sigma^2 = \iint (F(x \wedge y) - F(x)F(y)) dx dy = \text{Var}(Z)$$

by Hoeffding's covariance identity — the CLT for the sample mean.

3. Mean excess function

bounded bilinear + Lemma 1.10(2) + Theorem 1.19

$Z \geq 0$, $\mathbb{E}Z^2 < \infty$, $e(u) = \mathbb{E}[(Z - u)^+]/S(u)$ on $[0, \tau]$ with $S(\tau) > 0$.

Theorem. With $A(u) = \mathbb{E}[(Z - u)^+]$, $A_n(u) = n^{-1} \sum_j (Z_j - u)^+$, $S_n(u) = n^{-1} \sum_j \mathbf{1}\{Z_j > u\}$ and $\hat{e} = A_n/S_n$,

$$\sqrt{n}(\hat{e} - e) \rightsquigarrow \frac{\mathbb{G}_1 - e\mathbb{G}_2}{S} \quad \text{in } \ell^\infty([0, \tau]),$$

centred Gaussian with Cov given in [7].

Proof. [1] $e = \phi_{\text{prod}}(A, \phi_{\text{rec}}(S))$, where ϕ_{rec} is the pointwise map from $\varphi(x) = 1/x$ and $\phi_{\text{prod}}(f, g) = fg$. Two stages. [2] Only the denominator can fail: the bad set of φ' is $\{0\}$. S is non-increasing, so $\inf_{[0, \tau]} S = S(\tau) > 0$; put $\delta = \frac{1}{2}S(\tau)$ and $D_\phi = \ell^\infty([0, \tau]) \times \{h : h([0, \tau]) \subseteq (\delta, \infty)\}$, open. The numerator is unconstrained — cutting it would be an over-correction.

[3] On (δ, ∞) : $|\varphi'(x)| = x^{-2} \leq \delta^{-2}$ and $|\varphi''(x)| = 2x^{-3} \leq 2\delta^{-3}$, so Lemma 1.10(2) gives $\phi'_{\text{rec}}S(h_2) = -h_2/S^2$. The product map is bounded bilinear ($\|fg\|_\infty \leq \|f\|_\infty\|g\|_\infty$, cross term $O(t_n^2)$), hence Hadamard with derivative $(h_1, k) \mapsto h_1 \cdot g + f \cdot k$ at (f, g) . Composing at $(f, g) = (A, 1/S)$ with $k = -h_2/S^2$:

$$e'(h_1, h_2) = \frac{h_1}{S} - \frac{Ah_2}{S^2} = \frac{h_1 - e h_2}{S}.$$

[4] S_n is non-increasing, so $\inf_{[0, \tau]} S_n = S_n(\tau) \rightarrow S(\tau) = 2\delta$ a.s.; hence $\mathbb{P}((A_n, S_n) \in D_\phi) \rightarrow 1$.

[5] *Joint* convergence is required, since two processes enter the map. Let $\mathcal{F}_1 = \{z \mapsto (z - u)^+ : u \in [0, \tau]\}$ — Lipschitz in u with envelope $Z + \tau \in L^2$, so bracketing numbers of order ε^{-1} — and $\mathcal{F}_2 = \{z \mapsto \mathbf{1}\{z > u\}\}$, a VC class. Each is Donsker; a finite union of Donsker classes is Donsker, so Theorem 1.19 gives $\sqrt{n}(A_n - A, S_n - S) \rightsquigarrow (\mathbb{G}_1, \mathbb{G}_2)$ jointly in $\ell^\infty([0, \tau])^2$, tight.

[6] Theorem 1.17: $\sqrt{n}(\hat{e} - e) \rightsquigarrow (\mathbb{G}_1 - e\mathbb{G}_2)/S$.

[7] Write the limit as $W(u) = \mathbb{G}(\psi_u)$ with $\psi_u(z) = S(u)^{-1}[(z - u)^+ - e(u)\mathbf{1}\{z > u\}]$. Since $A = eS$ we have $\mathbb{E}\psi_u(Z) = 0$, so

$$\text{Cov}(W(u), W(v)) = \frac{\mathbb{E}[(Z - u)^+ - e(u)\mathbf{1}\{Z > u\}][(Z - v)^+ - e(v)\mathbf{1}\{Z > v\}]}{S(u)S(v)}. \quad \square$$

Check. At $u = v$ the numerator is $\mathbb{E}[\mathbf{1}\{Z > u\}(Z - u - e(u))^2] = S(u) \text{Var}(Z | Z > u)$, since $\mathbb{E}[Z - u | Z > u] = e(u)$. So $\text{Var } W(u) = \text{Var}(Z | Z > u)/S(u)$ — finite exactly when $\mathbb{E}Z^2 < \infty$, which is where that hypothesis is used. Marginal convergence of A_n and S_n separately would *not* have sufficed in [5].

4. Trimmed Lorenz curve

Lemma 1.10(1)(i) + Proposition 1.2(1), (2), (3)

$Z \geq 0$, $0 < \mu < \infty$, f continuous and > 0 on $[q_\delta, q_{1-\delta}]$; $L_\delta(p) = \mu_\delta^{-1} \int_\delta^p q_u du$, $\mu_\delta = \int_\delta^{1-\delta} q_u du$.

Theorem. On $[\delta, 1 - \delta]$,

$$\sqrt{n}(\hat{L}_\delta - L_\delta)(p) \rightsquigarrow \mathbb{L}(p) = \frac{1}{\mu_\delta} \left[\int_\delta^p G(u) du - L_\delta(p) \int_\delta^{1-\delta} G(u) du \right], \quad G = -\frac{B}{f \circ q}.$$

Proof. [1] $L_\delta = \phi_3 \circ \phi_2 \circ \phi_1$ with $\phi_1(F) = q$, $\phi_2(q) = (\int_\delta^{(\cdot)} q_u du, \int_\delta^{1-\delta} q_u du)$, $\phi_3(x, y) = x/y$.

[2] A single trimming level $\delta \in (0, \frac{1}{2})$, fixed once, is a stage-1 requirement: it makes $[\delta, 1 - \delta]$ compact in $(0, 1)$ as Lemma 1.10(1)(i) and Theorem 1.23 demand, and it is where $f > 0$ is assumed. Stage 2 needs nothing. Stage 3 needs only $y \neq 0$, an open condition, satisfied since $\mu_\delta > 0$ because $Z \geq 0$ has positive mean. Do not introduce a second trimming level.

[3] $\phi'_{1F}(h) = -(h/f) \circ F^{-1}$ tangentially to $C[a, b]$; ϕ_2 is bounded linear, hence its own derivative (Proposition 1.2(1), (2)); ϕ_3 is differentiable wherever $y \neq 0$ with $\phi'_3(a, b) = a/y - xb/y^2$. Chain rule, evaluated at $x = \int_\delta^p q$, $y = \mu_\delta$, $a = \int_\delta^p h_u du$, $b = \int_\delta^{1-\delta} h_u du$:

$$L'(h)(p) = \frac{1}{\mu_\delta} \int_\delta^p h_u du - \frac{L_\delta(p)}{\mu_\delta} \int_\delta^{1-\delta} h_u du.$$

Continuity is preserved at every stage, so the tangency conditions chain.

[4] $F_{(n)}$ is always a df, and $\hat{\mu}_\delta \rightarrow \mu_\delta \neq 0$ in probability, so the pair lies in the stage-3 domain with probability tending to one.

[5] Theorem 1.23: $\sqrt{n}(\hat{q} - q) \rightsquigarrow G = -B/(f \circ q)$ in $\ell^\infty([\delta, 1 - \delta])$, with continuous paths.

[6]–[7] Theorem 1.17 gives $\mathbb{L}$ as stated; it is a bounded linear functional of B , hence a centred Gaussian process, with covariance obtained by expanding the two integrals against $\text{Cov}(B(u), B(v)) = u \wedge v - uv$. $\square$

Check. $L_\delta(1 - \delta) = 1$ identically, so the limit must vanish there — and indeed $\mathbb{L}(1 - \delta) = \mu_\delta^{-1}[\int_\delta^{1-\delta} G - 1 \cdot \int_\delta^{1-\delta} G] = 0$. The trimmed Gini coefficient $1 - 2(1 - 2\delta)^{-1} \int L_\delta(p) dp$ is one further bounded linear map, so it costs one citation and no new conditions.

5. Expected shortfall

bounded linear map + Theorem 1.23

$\text{ES}_{\alpha, \delta} = (1 - \alpha - \delta)^{-1} \int_\alpha^{1-\delta} q_u du$.

Theorem. For $0 < \alpha < 1$, $0 < \delta < 1 - \alpha$, and f continuous and positive on $[q_\alpha, q_{1-\delta}]$,

$$\sqrt{n}(\widehat{\text{ES}} - \text{ES}) \rightsquigarrow N\left(0, (1 - \alpha - \delta)^{-2} \iint \frac{u \wedge v - uv}{f(q_u)f(q_v)} du dv\right).$$

Proof. [1] One stage: $\phi(q) = (1 - \alpha - \delta)^{-1} \int_\alpha^{1-\delta} q_u du$ on $\ell^\infty([\alpha, 1 - \delta])$.

[2] None. ϕ is linear, and $|\phi(q)| \leq (1 - \alpha - \delta)^{-1} (1 - \alpha - \delta) \|q\|_\infty = \|q\|_\infty$, so it is continuous on the whole space. There is no bad set, no value cut and no δ to choose. (The δ in the definition is a trimming level, not a domain slack.)

[3] Directly from Definition 1.12: the difference quotient $t_n^{-1}[\phi(q + t_n h_n) - \phi(q)] = \phi(h_n) \rightarrow \phi(h)$ by continuity. So $\phi'_q = \phi$, tangentially to all of $\ell^\infty([\alpha, 1 - \delta])$.

[4] $\hat{q} \in \ell^\infty([\alpha, 1 - \delta])$ always. Nothing to check.

[5] Theorem 1.23 on $[\alpha, 1 - \delta]$, compact in $(0, 1)$: $\sqrt{n}(\hat{q} - q) \rightsquigarrow G = -B/(f \circ q)$. This is the only place assumptions on f enter.

[6] Theorem 1.17: $\sqrt{n}(\widehat{\text{ES}} - \text{ES}) \rightsquigarrow (1 - \alpha - \delta)^{-1} \int_\alpha^{1-\delta} G(u) du$, a centred normal.

[7] Variance = $(1 - \alpha - \delta)^{-2} \iint \text{Cov}(B(u), B(v)) / (f(q_u)f(q_v)) du dv$, as stated. $\square$

Check. Letting $\alpha \rightarrow 0$, $\delta \rightarrow 0$ and substituting $u = F(x)$ reduces the variance to $\text{Var}(Z)$, as in Example 2. If you found yourself choosing an interval (a, b) here, you misread the map: ask “is it linear?” before anything else.

6. Kaplan–Meier product integral

Lemma 1.12

$\hat{S} = \prod_{[0, \cdot]} (1 - d\hat{\Lambda})$ with $\sqrt{n}(\hat{\Lambda} - \Lambda) \rightsquigarrow \mathbb{G}$ on $(0, b]$.

Theorem. Let Λ be continuous with $\Lambda(b) < \infty$ and $\sqrt{n}(\hat{\Lambda} - \Lambda) \rightsquigarrow \mathbb{G}$ in $D(0, b]$, $\hat{\Lambda}$ non-decreasing. Then

$$\sqrt{n}(\hat{S} - S) \rightsquigarrow -S \cdot \mathbb{G}, \quad \text{Cov}(s, t) = S(s)S(t) \text{Cov}(\mathbb{G}(s), \mathbb{G}(t)).$$

Proof. [1] $\hat{S} = \phi(-\hat{\Lambda})$, where ϕ is the product integral $F \mapsto \prod(1 + dF)$ of Lemma 1.12, precomposed with the bounded linear map $\Lambda \mapsto -\Lambda$ (its own derivative, Proposition 1.2(1)).

[2] The constraint is on *oscillation*, not on values: the product integral is a limit over partitions, so Lemma 1.12 is stated on $BV_M(0, b]$. Fix $M > \Lambda(b)$, say $M = 2\Lambda(b)$. Note BV_M is **not open** in $\|\cdot\|_\infty$ — a sup-norm-small perturbation can have unbounded variation — so the requirement $\theta + t_n h_n \in D_\phi$ genuinely restricts the admissible directions, and the tangential set is the set of directions of bounded variation. This must be stated, not glossed.

[3] Lemma 1.12 gives $\phi'_F(h)(t) = \int_{(0, t]} \phi(F)(0, u-) \phi(F)(u, t] dh(u)$. With $F = -\Lambda$ and Λ continuous, $\phi(F)(0, u-) = S(u)$ and $\phi(F)(u, t] = S(t)/S(u)$, so the integrand is the constant $S(t)$ and

$$\phi'_{-\Lambda}(h)(t) = S(t) \int_{(0, t]} dh(u) = S(t)h(t).$$

Composing with $\Lambda \mapsto -\Lambda$ replaces h by $-h$: the derivative of $\Lambda \mapsto \hat{S}$ is $h \mapsto -S(t)h(t)$.

[4] $\hat{\Lambda}$ is non-decreasing, so $\|\hat{\Lambda}\|_{TV(0, b]} = \hat{\Lambda}(b) \rightarrow \Lambda(b) = M/2 < M$ in probability. Check the total variation, not the supremum — this is the step most often done wrongly.

[5] Whatever supplies $\sqrt{n}(\hat{\Lambda} - \Lambda) \rightsquigarrow \mathbb{G}$: Theorem 1.22 composed with Example 1 in the uncensored case, or a martingale CLT under censoring. The limit must have paths of bounded variation, or be handled through the Duhamel equation (Lemma 1.14).

[6]–[7] Theorem 1.17: $\sqrt{n}(\hat{S} - S) \rightsquigarrow -S\mathbb{G}$, with the stated covariance by bilinearity. $\square$

Check. Sign bookkeeping first: the lemma is stated for $\prod(1 + dF)$, so survival functions enter as $F = -\Lambda$; getting this wrong flips the whole limit. Now specialise to no censoring, $\hat{\Lambda} = -\log(1 - F_{(n)})$. Then $\hat{S} = \exp(-\hat{\Lambda}) = 1 - F_{(n)}$, and by Example 1, $\mathbb{G} = (B \circ F)/(1 - F)$, so

$$-S\mathbb{G} = -(1 - F) \cdot \frac{B \circ F}{1 - F} = -B \circ F,$$

which is exactly $\sqrt{n}(\hat{S} - S) = -\sqrt{n}(F_{(n)} - F)$ from Theorem 1.22. The two routes agree.

7. Empirical copula

Lemma 1.10(1),(3) + Theorem 1.16

$$C_{(n)}(u, v) = H_{(n)}([F_1^{(n)}]^{-1}(u), [F_2^{(n)}]^{-1}(v)).$$

Theorem. Let H have continuous margins F_1, F_2 with f_i continuous and positive on $[a, b], [c, d]$, and let $\partial H/\partial x, \partial H/\partial y$ be continuous there. Then $\sqrt{n}(C_{(n)} - C) \rightsquigarrow \mathbb{G}_C$ with

$$\mathbb{G}_C(u, v) = \mathbb{B}_C(u, v) - \partial_1 C(u, v) \mathbb{B}_C(u, 1) - \partial_2 C(u, v) \mathbb{B}_C(1, v),$$

where $\mathbb{B}_C$ is the C -Brownian bridge, $\mathbb{B}_C(u, v) = \mathbb{G}_H(F_1^{-1}(u), F_2^{-1}(v))$.

Proof. [1] Three catalogue maps in sequence: $\phi(H) = (H, H(\cdot, \infty), H(\infty, \cdot))$, linear; $\psi(H, F_1, F_2) = (H, F_1^{\leftarrow}, F_2^{\leftarrow})$, the inverse in two slots; $\nu(H, q_1, q_2)(x, y) = H(q_1(x), q_2(y))$, the composition map.

[2] Stage ψ : two index intervals $[a, b], [c, d]$, each an ε -enlargement of $[F_i^{-1}(p), F_i^{-1}(q)]$, with f_i continuous and positive — exactly as in the univariate inverse lemma, done twice. Stage ν : the genuinely new condition is that the *outer* function be uniformly Fréchet differentiable with uniformly bounded derivative, which continuity of $\partial H/\partial x, \partial H/\partial y$ on the compact rectangle supplies. There is no value cut anywhere.

[3] $\phi' = \phi$; $\psi'(a, b, c) = (a, -(b/f_1) \circ F_1^{-1}, -(c/f_2) \circ F_2^{-1})$ by Lemma 1.10(1)(i); and by Lemma 1.10(3),

$$\nu'(a, b, c) = a \circ (F_1^{-1}, F_2^{-1}) + \langle (\partial_x H, \partial_y H) \circ (F_1^{-1}, F_2^{-1}), (b, c) \rangle.$$

Theorem 1.16 composes them. Tangency must be checked stage by stage: D_0 for ν is $\ell^\infty \times UC$, and the image of the previous derivative consists of continuous functions on a compact rectangle, hence uniformly continuous. Using $\partial_x H(F_1^{-1}(u), F_2^{-1}(v))/f_1(F_1^{-1}(u)) = \partial_1 C(u, v)$ gives the displayed form.

[4] $H_{(n)}$ and its margins are dfs for every n ; nothing to check.

[5] Theorem 1.22, bivariate version: $\sqrt{n}(H_{(n)} - H) \rightsquigarrow \mathbb{G}_H$, tight with continuous paths under continuous margins.

[6]–[7] Theorem 1.17 gives $\mathbb{G}_C$; its covariance follows from that of $\mathbb{B}_C$, namely $C(u \wedge u', v \wedge v') - C(u, v)C(u', v')$, expanded through the three terms. $\square$

Check. The three terms say: you pay once for the joint df and once for each estimated margin. Consistency at the boundary: $C(u, 1) = u$ is deterministic, so $\mathbb{G}_C(u, 1)$ must vanish — and indeed $\partial_1 C(u, 1) = 1, \mathbb{B}_C(1, 1) = 0$, giving $\mathbb{B}_C(u, 1) - 1 \cdot \mathbb{B}_C(u, 1) - \partial_2 C(u, 1) \cdot 0 = 0$.

$$8. G(t) = \int_{(-\infty, t]} \psi(F(s-)) dF(s)$$

Lemma 1.11 + diagonal map

$\psi \in C^2[0, 1]$; with $\psi(x) = x$ this is $\mathbb{P}(Z_1 \leq Z_2 \leq t)$.

Theorem. For $\psi \in C^2[0, 1]$, $\sqrt{n}(G_{(n)} - G) \rightsquigarrow G'_F(B \circ F)$ in $D[-\infty, \infty]$, where

$$G'_F(h)(t) = [h\psi(F)]_{-\infty}^t - \int_{(-\infty, t]} h(s-) d\psi(F(s)) + \int_{(-\infty, t]} \psi'(F(s)) h(s) dF(s).$$

Proof. [1] Lemma 1.11 is stated for a *pair* (F_1, F_2) , but both slots here hold the same F . Precompose with the diagonal $\Delta(F) = (F, F)$, which is linear and bounded, hence free: $G = g \circ \Delta$ with g the map of Lemma 1.11.

[2] Two constraints of different kinds, both free here. *Variation*: a df is non-decreasing from 0 to 1, so its total variation is exactly 1 and $M = 1$ works in $BV_M \times BV_M$ — never needs verifying. *Value*: ψ must be C^2 on an interval containing the range of the first argument, and that range lies in $[0, 1]$, so $\psi \in C^2[0, 1]$ suffices. No index truncation is needed.

[3] $\Delta' = \Delta$, so Lemma 1.11 applied with both partials and summed gives the three displayed terms: a boundary term, an integration-by-parts term from perturbing the integrator, and a chain-rule term from perturbing the integrand.

[4] Automatic, by [2] — but say so in one sentence rather than skipping it.

[5] Theorem 1.22: $\sqrt{n}(F_{(n)} - F) \rightsquigarrow B \circ F$, tight in $D[-\infty, \infty]$, with paths of bounded variation handled by the BV tangency of Lemma 1.11.

[6]–[7] Theorem 1.17: replace h by $B \circ F$ throughout. The limit is centred Gaussian; its covariance follows by expanding the three terms pairwise against $\text{Cov}(B(F(s)), B(F(t))) = F(s \wedge t) - F(s)F(t)$. $\square$

Check. Take $\psi(x) = x$ and F continuous, so $G(t) = \int_{-\infty}^t F dF = \frac{1}{2}F(t)^2$. The three terms give

$$h(t)F(t) - \int_{(-\infty, t]} h dF + \int_{(-\infty, t]} h dF = h(t)F(t),$$

the last two cancelling, which is exactly the derivative of $F \mapsto \frac{1}{2}F^2$. If your answer has fewer than three terms you differentiated only one slot.

9. Modulus of the empirical characteristic function

finite-dimensional differentiability

$\varphi_n(t) = n^{-1} \sum_j e^{itZ_j}$; asymptotics of $|\varphi_n(t_0)|$ for fixed t_0 .

Theorem. If $\varphi(t_0) \neq 0$ then $\sqrt{n}(|\varphi_n(t_0)| - |\varphi(t_0)|) \rightsquigarrow N(0, \sigma^2)$ with

$$\sigma^2 = \frac{1}{2} + \frac{1}{2} \frac{\text{Re}(\varphi(2t_0)\overline{\varphi(t_0)}^2)}{|\varphi(t_0)|^2} - |\varphi(t_0)|^2.$$

Proof. [1] $\phi = |\cdot| \circ \text{ev}_{t_0}$, with ev_{t_0} bounded linear on $\ell^\infty([-T, T])$. After evaluation the problem lives in $\mathbb{C} \cong \mathbb{R}^2$, where Hadamard differentiability is *equivalent* to ordinary total differentiability.

[2] $|\cdot|$ has bounded difference quotients everywhere but no derivative at 0, and the derivative is not continuous there: boundedness is not the only hypothesis. So assume $\varphi(t_0) \neq 0$, put $\delta = \frac{1}{2}|\varphi(t_0)|$ and work on the open set $\{z \in \mathbb{C} : |z| > \delta\}$, where $|\cdot|$ is smooth. This is a *kink* excluded, not a blow-up.

[3] On $\mathbb{R}^2$, $\nabla|z| = z/|z|$, so the derivative in direction h is $\text{Re}(\bar{z}h)/|z|$; chain rule with ev_{t_0} .

[4] $|\varphi_n(t_0)| \rightarrow |\varphi(t_0)| = 2\delta$ a.s. by the LLN, so $\mathbb{P}(\varphi_n(t_0) \in D_\phi) \rightarrow 1$.

[5] $\{e^{it} : t \in [-T, T]\}$ is Donsker (Lipschitz in t , bounded envelope), giving $\sqrt{n}(\varphi_n - \varphi) \rightsquigarrow G$ with $\text{Cov}(G(s), G(t)) = \varphi(s-t) - \varphi(s)\varphi(-t)$; for a single t_0 the bivariate CLT for $(\cos t_0 Z, \sin t_0 Z)$ suffices.

[6] Theorem 1.17: the limit is $\text{Re}(\bar{u} G(t_0))$ with $u = \varphi(t_0)/|\varphi(t_0)|$, centred normal.

[7] Write $\varphi(t_0) = |\varphi(t_0)|e^{i\theta}$, so $\bar{u} = e^{-i\theta}$ and $W := \text{Re}(\bar{u}e^{it_0 Z}) = \cos(t_0 Z - \theta)$. Then $\mathbb{E}W = \text{Re}(e^{-i\theta}\varphi(t_0)) = |\varphi(t_0)|$ and

$$\mathbb{E}W^2 = \mathbb{E}\cos^2(t_0 Z - \theta) = \frac{1}{2} + \frac{1}{2} \text{Re}(e^{-2i\theta}\varphi(2t_0)), \quad e^{-2i\theta} = \frac{\overline{\varphi(t_0)}^2}{|\varphi(t_0)|^2},$$

and $\sigma^2 = \mathbb{E}W^2 - (\mathbb{E}W)^2$ is the stated expression. $\square$

Check. If Z is symmetric then φ is real and $\sigma^2 = \frac{1}{2} + \frac{1}{2}\varphi(2t_0) - \varphi(t_0)^2 = \text{Var}(\cos t_0 Z)$, correct because $|\varphi_n|$ is then the real part.

10. Relative distribution and the ROC curve

Lemma 1.10(1) + Lemma 1.10(3)

Independent samples from F (size n) and G (size m); $R(u) = G(F^{\leftarrow}(u))$ on $[p, q] \subset (0, 1)$.

Theorem. Let f be continuous and positive on $[a, b] = [F^{-1}(p) - \varepsilon, F^{-1}(q) + \varepsilon]$, let $G \in C^1$ with g bounded and uniformly continuous there, and let $n/(n+m) \rightarrow \lambda \in (0, 1)$. Then, with $k_{n,m} = \sqrt{nm/(n+m)}$,

$$k_{n,m}(\hat{R} - R)(u) \rightsquigarrow \sqrt{\lambda} B_2(R(u)) - \sqrt{1-\lambda} \frac{g(q_u)}{f(q_u)} B_1(u),$$

a sum of two independent centred Gaussian terms, with variance $\lambda R(u)(1-R(u)) + (1-\lambda)(g/f)^2(q_u)u(1-u)$.

Proof. [1] $R = \nu(G, \phi_{\text{inv}}(F))$, with ν the composition map of Lemma 1.10(3). Two stages, two different reasons to restrict.

[2] Stage 1 (inversion): $[p, q] \subset (0, 1)$ and f continuous, positive on $[a, b]$. Stage 2 (composition): G uniformly Fréchet differentiable over the range of $F^{\leftarrow}$, which for a real function means $G \in C^1$ with g bounded and uniformly continuous. Nothing constrains the *values* of either df: there is no (a, b) value cut and no δ .

[3] Lemma 1.10(1)(i): $\phi'_{\text{inv}}(h_1) = -(h_1/f) \circ F^{-1}$, tangentially to $C[a, b]$. Lemma 1.10(3) then gives

$$R'(h_1, h_2)(u) = h_2(F^{\leftarrow}(u)) - \frac{g(F^{\leftarrow}(u))}{f(F^{\leftarrow}(u))} h_1(F^{\leftarrow}(u)).$$

Tangency: the image of $C[a, b]$ under the first derivative is continuous on a compact interval, hence uniformly continuous, so it lies in $D_0 = \ell^\infty \times UC$.

[4] Both empirical dfs are dfs. Note that $G_{(m)}$ is *not* C^1 — but the smoothness hypothesis is on the true G , the point at which we differentiate, not on the estimator. Worth one sentence.

[5] Theorem 1.22 twice, plus independence: the two empirical processes are independent and each tight with continuous limit, so the pair converges jointly to $(B_1 \circ F, B_2 \circ G)$ with independent components. No Donsker class for the pair is needed — independence gives the joint limit for free.

[6] Theorem 1.17, using $F(q_u) = u$ and $G(q_u) = R(u)$, after multiplying by $k_{n,m}$ (so that $k_{n,m}/\sqrt{m} \rightarrow \sqrt{\lambda}$ and $k_{n,m}/\sqrt{n} \rightarrow \sqrt{1-\lambda}$), gives the displayed limit.

[7] The two terms are independent, so the variances add: $\lambda(R(u) \wedge R(u) - R(u)^2) + (1-\lambda)(g/f)^2(q_u)(u - u^2)$, as stated. $\square$

Check. If $G = F$ then $R(u) = u$ and $g = f$, so the variance is $\lambda u(1-u) + (1-\lambda)u(1-u) = u(1-u)$: a standard Brownian bridge, as it must be when the two samples come from the same law. The decomposition also explains why ROC estimation is hard in the tails — as $u \rightarrow 0$ or 1 the factor g/f evaluated at an extreme quantile need not stay bounded, and the second term dominates.

Across all ten. Only Examples 1, 3 and 9 require a δ ; in 2, 4, 5, 7, 10 the localisation is in the index set and in 6, 8 it is a variation bound. Only Examples 3, 7 and 10 need joint convergence, and only 3 pays for it with a Donsker class — 7 gets it from a bivariate theorem, 10 from independence. Every proof spends most of its length on Steps 2–4; Steps 5–7 are one citation and one bilinear expansion each. That ratio is the point.